

AYUSH KUMAR

+91-6203275940 ✦ ayushhoff@gmail.com ✦ Bengaluru, Karnataka, India

[Portfolio Website](#) ✦ [GitHub Profile](#) ✦ [LinkedIn Profile](#)

OBJECTIVE

Third-year CS student at RVITM building production-grade AI systems. Looking for internships in AI/ML or backend engineering where I can contribute to real infrastructure, not just coursework

EDUCATION

RV Institute of Technology and Management

Bachelor of Engineering in Computer Science & Engineering

Expected 2027

CGPA till 5th Sem: 9.03

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, C++, Java, C
Backend	FastAPI, Node.js, Express, REST APIs, WebSockets, Async Processing, Pydantic, JWT, Celery
Frontend	Next.js, React, Tailwind CSS, Vite, D3.js
Databases	PostgreSQL, Redis, MongoDB, MySQL
Cloud & DevOps	AWS, Docker, GitHub Actions, CI/CD Pipelines, Linux, Git
AI/ML	PyTorch, HF Transformers, scikit-learn, LangChain, OpenAI API, Anthropic API, NumPy, Pandas
Concepts	Distributed Systems, Event-Driven Architecture, LLM Tool Calling, MCP Protocol, RAG Pipelines

PROJECTS – REAL-TIME SYSTEMS & ML PLATFORMS

ENGRAM — Self-Hostable AI Memory Layer for Cross-LLM Context Sync

May 2026

AI memory layer that retrieves, injects, extracts, and syncs durable user context across LLM clients.

[Live Demo](#) | [GitHub](#)

- Built a FastAPI proxy that injects relevant pgvector memories into LLM requests before provider forwarding.
- Added async extraction to capture durable user facts after conversations and deduplicate repeated memories.
- Implemented semantic search, confidence scoring, access tracking, retrieval logs, and per-user memory settings.
- Developed a TypeScript MCP server for Claude Desktop, VS Code Agent Mode, Cursor-style clients, and agent tools.
- Shipped a Next.js dashboard to inspect, search, edit, delete, and debug memories, logs, keys, and sync state.

SENTINEL - Multi-Agent Market Microstructure Simulator with Predictive Analytics

Ongoing

Advanced market simulation platform for liquidity crisis prediction and large order detection.

[Live Demo](#) | [GitHub](#)

- Built a FastAPI + Next.js simulator with 40+ trading agents, WebSocket streams, and live order-book analytics.
- Modeled market makers, HFT, retail, noise, informed, and institutional agents under latency and stress scenarios.
- Developed liquidity-shock alerts with 60–90s lead time using order-book depth, spread, imbalance, and volatility signals.
- Implemented TWAP/VWAP/Iceberg detection to identify hidden institutional order flow and execution patterns.
- Designed a terminal-style dashboard for price, spread, depth, PnL, agent activity, liquidity risk, and event monitoring.
- Tools: Python, FastAPI, TypeScript, Next.js, WebSockets, XGBoost, scikit-learn, NumPy, LOBSTER-style data.

EQUITYFLOW - Real-Time Paper Trading for Stocks, F&O, and Commodities

Feb 2026 – Mar 2026

Real-time paper trading simulator for stocks, F&O, and commodities.

[Live Demo](#) | [GitHub](#)

- Built a real-time paper trading simulator for stocks, F&O, and commodities with live broker data integration.
- Improved price freshness using SSE streams, shorter cache TTLs, and faster polling fallback update logic.
- Implemented portfolio tracking for holdings, orders, average price, realized P&L, unrealized P&L, and cash.
- Added F&O contract resolution for 40k+ instruments using indexed symbol, expiry, and segment metadata.
- Designed a compact trading-desk UI with charts, watchlists, order tickets, diagnostics, and provider controls.
- Tools: Next.js, React, TypeScript, FastAPI, Python, Tailwind CSS, REST APIs, WebSockets, broker APIs.

Text-Based Parkinson's Disease Screening using Transformers

Jul 2025

Multimodal ML pipeline for clinical classification using transformers and ensemble models.

[GitHub](#)

- Developed an NLP-based diagnostic pipeline using transformer models on 42k patient records, deploying an inference API with Flask achieving 92% accuracy.
- Fine-tuned PubMedBERT, BioGPT, Clinical-T5 on 42K+ patient records.
- Built reproducible training + evaluation pipelines with cross-validation and ablation studies.
- Implemented a production-ready inference API using Flask for clinical prediction services.
- Tools: PyTorch, Transformers, LightGBM, Flask, CUDA

OCCASIO - Event Management & Booking Platform

Oct 2025 – Nov 2025

Event platform for publishing, payments, QR tickets, check-in, analytics, teams, and certificates.

[Live Demo](#) | [GitHub](#)

- Built a React + Vite frontend for event discovery, registration, admin dashboards, and scanner check-in.
- Developed Node.js + Express + Prisma APIs for event CRUD, auth, analytics, teams, tickets, and wallet features.

- Integrated Razorpay/PhonePe payments with secure QR tickets, PDF generation, email delivery, and downloads.
- Added organizer tools for forms, ticket styling, discounts, polls, reminders, reviews, teams, and certificates.
- Prepared deployment with PostgreSQL, BullMQ/Redis queues, cloud storage, PWA support, and separated APIs.

OPEN SOURCE CONTRIBUTIONS

CDLI (Cuneiform Digital Library Initiative)

Mar 2026

Contributed 10 merge requests to the CDLI GitLab repository, 5 merged, improving repository tooling and data processing workflows.

Gemini CLI (Google)

Mar 2026

Contributed 4+ pull requests improving CLI tooling, including WSL shell execution routing fixes and trusted tool configuration enhancements.

Processing Foundation

Dec 2023

Contributed pull request fixing UI bug in Processing website repository, resolving incorrect search placeholder behavior.

ACHIEVEMENTS

Codeforces Contest

Dec 2024

Ranked top 20% in global Codeforces contest (52k+ participants) worldwide.

Competitive Programming

Ongoing

Solved 300+ algorithmic problems across Codeforces, LeetCode, CodeChef, and AtCoder.

Research Publications

2025

Published two ML preprints on Zenodo: *Optimizing Soybean Production: A Data-Driven Machine Learning Framework for Disease Prediction and Yield Improvement* and *Context is All You Need: Why Transformers Outperform Traditional ML Models in Tweet Sentiment Classification*.

POSITIONS OF RESPONSIBILITY

Technical Head

2025 – Present

Robotics, Web Dev, Cybersecurity, Coding and Entrepreneurship Club

Events and Website Head

2025 – Present

Core Organizing Committee, National Level Techfest Robofiesta 2025

Core Team Member

2025 – Present

Robotics, Web Dev, Cybersecurity, Coding, GDG and Entrepreneurship Club